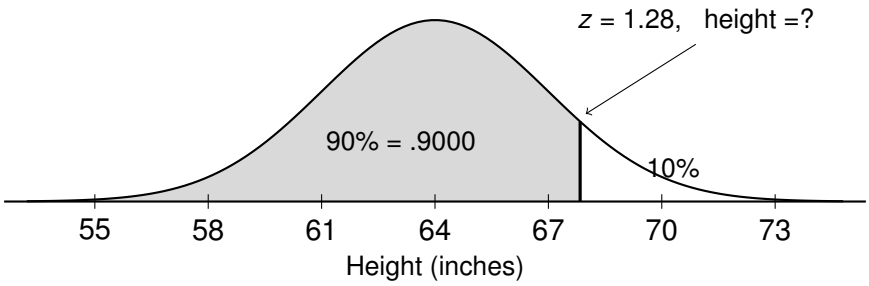
Normal Calculations in Reverse KEY

Name _____

In 2.5, you knew the value and found the percent. Now you know the percent and find the value.

Example: Heights of adult women: mean 64 inches, SD 3 inches. How tall is a woman at the **90th percentile**?



Steps	**Example: 90th percentile**
1. Draw the curve. Shade the area to the **left**.	90% is to the left. The area is .9000.
2. Look **inside** the table for the area closest to it.	The closest area is .8997.
3. Read the z-score from the row and the column.	Row 1.2, column .08: $z = 1.28$
4. Find the value: $x = \mu + z\sigma$	$64 + 1.28 \times 3 = 64 + 3.84 = 67.84$

A woman at the 90th percentile is about **67.8 inches** tall.

"Top" and "bottom" questions

The **bottom 10%**	area to the left is .10
The **top 10%**	area to the left is $1 - .10 = .90$

Example: Below what height are the shortest 10% of women?

Area .1000. Closest in the table: .1003 at $z = -1.28$. $64 + (-1.28)(3) = 64 - 3.84 = 60.16$. About **60.2 inches**.

Gotcha: The table gives areas to the **left**. For "top" questions, subtract from 1 before you look in the table.

Use your z-table sheet for the problems.

Work with a partner. Compare answers when you finish each problem.

1. Use the z-table. Find the z-score at each percentile.

   (a) 80th percentile closest area .7995: $z = 0.84$

   (b) 30th percentile closest area .3015: $z = -0.52$

   (c) 97.5th percentile area .9750: $z = 1.96$

   (d) 25th percentile closest area .2514: $z = -0.67$

2. A test has scores with a normal distribution. The mean is 500 and the SD is 100.

   (a) What score is at the 90th percentile?
   $z = 1.28$. $500 + 1.28 \times 100 = 628$

   (b) What score is at the 30th percentile?
   $z = -0.52$. $500 + (-0.52)(100) = 448$

3. A phone battery lasts a mean of 20 hours with an SD of 2 hours. The times have a normal distribution. The company replaces any battery in the bottom 10%. Below what time does a battery get replaced?

   Area .10: $z = -1.28$. $20 + (-1.28)(2) = 20 - 2.56 = 17.44$. About 17.4 hours.

4. Apples have a mean weight of 150 grams and an SD of 10 grams. The weights have a normal distribution. The heaviest 15% are sold as "premium." How heavy must an apple be to be premium?

   Top 15%, so the area to the left is $1 - .15 = .85$. Closest: .8508 at $z = 1.04$. $150 + 1.04 \times 10 = 160.4$ grams.

5. Use the test from problem 2 (mean 500, SD 100). One question goes forward and one goes in reverse.

   (a) What percentile is a score of 640?

   $z = \dfrac{640 - 500}{100} = 1.4$. Table: .9192. About the 92nd percentile.

   (b) What score is at the 75th percentile?

   Closest area .7486 at $z = 0.67$. $500 + 0.67 \times 100 = 567$

6. **Challenge.** The heights of a type of plant have a normal distribution with a mean of 30 cm. 97.5% of the plants are shorter than 39.8 cm. Find the SD.

   97.5th percentile: $z = 1.96$. $30 + 1.96 \times \text{SD} = 39.8$, so $1.96 \times \text{SD} = 9.8$ and $\text{SD} = 9.8 \div 1.96 = 5$ cm.